A Mathematical Definition of LLM Outputs

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Abstract

Large Language Models (LLMs) are commonly interpreted as probabilistic next-token generators. However, this interpretation alone does not fully characterize the semantic nature of model outputs, especially under incomplete information. In this work, I propose a mathematical framework that defines LLM outputs as information-conditioned projections over latent semantic states. Instead of treating model outputs as deterministic truths, I formalize generation as an inference process conditioned on an evolving information filtration. I introduce the concept of an Information-Neutral Measure, under which LLM outputs can be interpreted analogously to conditional expectations. Under this framework, hallucination is not merely a system failure, but a natural consequence of incomplete information prior to oracle observation. This perspective unifies ideas from probabilistic inference, information theory, and generative AI into a common mathematical structure. My formulation provides a rigorous mathematical derivation showing how information completeness inherently bounds semantic uncertainty, providing a formal foundation for future work in generative AI theory.

1 Introduction

Modern Large Language Models (LLMs) have demonstrated remarkable capabilities across reasoning, coding, retrieval, and generation tasks. Existing interpretations typically model LLMs as autoregressive probabilistic systems:

$$P(y_t \mid y_{<t}, x) \tag{1}$$

where generation is framed as next-token prediction conditioned on prior context.

While operationally effective, this formulation does not fully capture the semantic uncertainty underlying model outputs. In particular, hallucinations reveal that LLM outputs cannot always be interpreted as deterministic representations of reality.

In this paper, I propose a higher-level mathematical interpretation of LLM outputs. I argue that LLM outputs are not deterministic truths, but information-conditioned projections over latent semantic states. I formalize this interpretation using an information-theoretic framework inspired by probability theory, stochastic processes, and financial asset pricing.

I define the central formulation as:

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) \tag{2}$$

where P_t denotes the generated output at time t, Ψ represents the latent semantic truth state, $\mathcal{F}_t$ denotes the available information filtration (σ-algebra stream), QI is an Information-Neutral Measure, and Π is an information-conditioned projection operator.

This framework suggests that hallucinations naturally emerge when the information filtration is insufficient to collapse uncertainty regarding the latent semantic state.

2 Information State Space and Core Definitions

I define an information state space by the triplet:

$$(\Omega, \mathcal{F}, Q_I) \tag{3}$$

where Ω is the semantic sample space, $\mathcal{F}$ is the global information σ-algebra, and Q_I is the Information-Neutral Measure.

I formally introduce the following core mathematical definitions mapped directly into LLM-centric terminology:

1. **Information (Context Stream $\mathcal{F}_t$):** The historical sequence of tokens, prompts, and retrieved knowledge blocks available up to step t. It acts as the filtration mapping the model's observable universe.

2. **Information Closure (Contextual Completeness $\mathcal{C}$):** The state where the sigma-algebra $\mathcal{F}_t$ contains sufficient entropic structure to uniquely resolve the true latent state Ψ, meaning no additional external prompt can alter the projection.

3. **Event (Semantic State Transition E):** A specific subset of the sample space Ω corresponding to a coherent semantic meaning or a targeted factual statement generated by the model.

4. **Event Contract (Generation Prompt Target $\mathcal{T}$):** A structured constraint or prompt requirement specifying the desired semantic properties of the output, analogous to a payoff structure depending on the realized semantic state.

5. **Result (Output Sequence / Final Token y^*):** The observable realization or factual ground truth that emerges once the model output is verified by an external oracle or finalized in text space.

6. **Oracle (Ground-Truth Verifier / Environment $\mathcal{O}$):** An external system, human evaluator, or environment that provides definitive verification of a generated statement, collapsing the probabilistic state into a deterministic result.

7. **Participant (Attention Sub-Layer / Agent Component $\mathcal{P}_i$):** The internal components of the system (such as individual attention heads or multi-agent nodes) that process distinct segments of the distributed training distribution.

8. **Belief Function (Predictive Belief / Logit Mapping $\mathcal{B}$):** The model's internal subjective probability distribution (represented via softmax logit space) over potential outputs prior to information collapse.

Unlike classical probabilistic language modeling, the semantic state Ψ is not assumed to be directly observable prior to oracle verification. Instead, generation occurs conditionally on partial information.

3 Mathematical Definition of LLM Outputs

I define LLM outputs as projections over latent semantic states conditioned on available information. The core definition is given by:

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) \tag{4}$$

This formulation generalizes traditional autoregressive token probabilities into semantic-state inference. Under this interpretation, components such as prompts, retrieved documents, model parameters, memory systems, tool outputs, and external environments all contribute to the filtration $\mathcal{F}_t$. Thus, generation is fundamentally an information aggregation process driven by the internal *Predictive Belief* $\mathcal{B}$.

3.1 Quantum Analogy: Superposition and State Collapse

To intuitively conceptualize this mechanism, the mathematical formulation $P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t)$ can be interpreted through the lens of quantum mechanics. Prior to text tokenization or external verification, an LLM's inner parameter space does not hold a singular deterministic truth; rather, it maintains a *semantic superposition state* over the latent universe Ψ.

This framework yields two profound physical analogies:

1. **Schrödinger's Cat in Latent Space:** Before the output token stream is decoded or evaluated by the Oracle ($\mathcal{O}$), the model's response is simultaneously "fact" and "hallucination" under the measure Q_I. The dynamic context stream $\mathcal{F}_t$ act as weak measurements that narrow down the density matrix of probability. The definitive verification by the Oracle represents a wave-function collapse, forcing the semantic superposition to resolve into a single, concrete reality y^*.

2. **Heisenberg's Semantic Uncertainty Principle:** An inherent epistemic limitation governs the generation process. If one attempts to perfectly pin down the localized token-level token sequence optimization (the immediate "momentum" of token prediction), the broader contextual semantic trajectory (the global "position" in the knowledge space Ψ) becomes highly volatile, leading to localized fluent coherence but global structural hallucination. Mathematically, the non-commutativity between the projection operator Π and the changing information filtration $\mathcal{F}_t$ limits the simultaneous resolution of local precision and global factual truth.

4 Mathematical Derivation of Generative Uncertainty

In this section, I provide the formal mathematical derivation demonstrating how information restriction dynamically governs the emergence of generative error without necessitating experimental hypotheses.

Let $\Psi \in L^2(\Omega, \mathcal{F}, Q_I)$ be a square-integrable latent semantic state representing the complete objective factual matrix. The filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ forms a completed, right-continuous filtration representing the progressive acquisition of context.

By defining the projection operator $\Pi_{QI}(\cdot \mid \mathcal{F}_t)$ as the orthogonal projection onto the closed subspace of L^2 that is $\mathcal{F}_t$-measurable, we can equivalently express the model's optimal information estimation as a conditional expectation under the information-neutral measure Q_I:

$$P_t = \Pi_{QI}(\Psi \mid \mathcal{F}_t) = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \tag{5}$$

By the law of total expectation, for any sequential information increase from s to t where $s \leq t$, the tower property holds:

$$\mathbb{E}_{Q_I}[P_t \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_t] \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[\Psi \mid \mathcal{F}_s] = P_s \tag{6}$$

This derivation formally proves that the stochastic sequence of LLM outputs $\{P_t\}_{t \geq 0}$ forms a **Martingale** with respect to the context stream filtration $\mathbb{F}$ under the measure Q_I.

To quantify the divergence from the Oracle reality Ψ^*, I define the semantic error variance (the hallucination distance square) at step t as:

$$\sigma_t^2 = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_t] \tag{7}$$

By applying the conditional variance decomposition (or law of total variance):

$$\mathbb{E}_{Q_I}[(\Psi - P_s)^2 \mid \mathcal{F}_s] = \mathbb{E}_{Q_I}[(\Psi - P_t)^2 \mid \mathcal{F}_s] + \mathbb{E}_{Q_I}[(P_t - P_s)^2 \mid \mathcal{F}_s] \tag{8}$$

Taking expectations on both sides yields the monotonic convergence structure:

$$\mathbb{E}_{Q_I}[\sigma_s^2] = \mathbb{E}_{Q_I}[\sigma_t^2] + \mathbb{E}_{Q_I}[(P_t - P_s)^2] \geq \mathbb{E}_{Q_I}[\sigma_t^2] \tag{9}$$

This mathematical derivation rigorously confirms that the expected factual error variance is strictly non-increasing over time as context expands ($s \leq t$).

5 Hallucination as Incomplete Information Projection

Traditional discussions often treat hallucination as an algorithmic or structural model defect. In contrast, I define hallucination mathematically as the condition:

$$\Pi_{QI}(\Psi \mid \mathcal{F}_t) \neq \Psi^* \tag{10}$$

where Ψ^* denotes the oracle-observed reality ($\mathcal{O}$).

Under this definition, hallucination emerges naturally when the available information is incomplete (lacking *Contextual Completeness*), the latent semantic state remains unresolved, or the projection operator is imperfectly calibrated. This interpretation resembles state collapse phenomena in probabilistic physical systems, where uncertainty persists prior to exact observation by the *Oracle*.

Corollary 1 (The Bound of Inevitable Hallucination)**.** *In any autoregressive generative system operating under an incomplete filtration ($\mathcal{F}_t \subset \mathcal{F}_{oracle}$), the projection operator $\Pi_{QI}(\Psi \mid \mathcal{F}_t)$ yields a non-zero semantic entropy. Consequently, absolute hallucination elimination is theoretically impossible prior to Information Closure ($\mathcal{C}$). Hallucination is an intrinsic epistemic property of the system, not an engineering defect.*

6 Implications

This framework suggests several broader implications for the field of artificial intelligence:

1. LLMs should not be interpreted as deterministic truth engines.
2. Uncertainty and hallucination are not accidental anomalies, but intrinsic properties of incomplete information systems before reaching *Information Closure*.

3. Future AI systems may benefit from explicitly modeling semantic uncertainty, information completeness, and oracle verification processes.

4. This perspective provides a potential bridge between generative AI, probabilistic inference, information theory, and stochastic mathematical frameworks.

7 Conclusion

I introduced a mathematical framework interpreting LLM outputs as information-conditioned projections over latent semantic states. Rather than viewing generation as deterministic truth production, my framework models LLMs as information aggregation systems operating under incomplete information. This perspective provides a formal interpretation of hallucination and a rigorous mathematical derivation proving how information completeness inherently governs output variance. I believe this formulation may offer a useful theoretical foundation for future work in generative AI, uncertainty modeling, and information-theoretic machine learning.